



姓名

实验报告

姓名

班级

组别

实验日期 2025.5.19

实验名称 电路的频率特性 实验指导老师

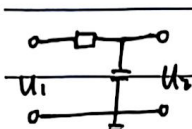
成绩

实验目的: 1. 测定无源线性电路的幅频特性.

2. 掌握低通、高通和带通网络的频率特性.

实验原理: • 网络函数. $|H| \rightarrow$ 幅频特性.

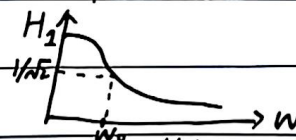
• 低通电路.



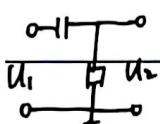
$$H(j\omega) = \frac{U_2}{U_1} = \frac{j\omega C}{R + j\omega C} = \frac{1}{1 + j\omega RC} \quad \text{令 } \omega_0 = \frac{1}{RC}, \text{ 则 } H(j\omega) = \frac{1}{1 + j\frac{\omega}{\omega_0}} = \frac{1}{\sqrt{1 + (\frac{\omega}{\omega_0})^2}} \angle -\tan^{-1}\frac{\omega}{\omega_0}$$

(截止频率) $|H(\omega)| = \frac{1}{\sqrt{1 + (\frac{\omega}{\omega_0})^2}}, \quad \varphi(\omega) = -\tan^{-1}\frac{\omega}{\omega_0}$

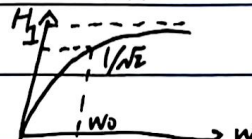
定义分贝(dB)作单位, $H(\omega)/dB = 20 \cdot \lg |H(j\omega)|$.



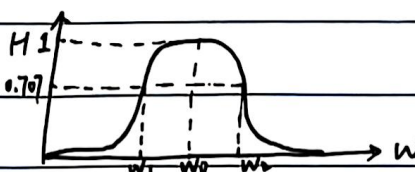
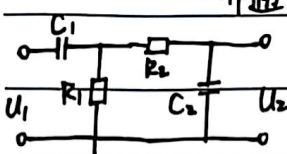
• 高通电路.



$$H(j\omega) = \frac{U_2}{U_1} = \frac{j\omega RC}{1 + j\omega RC} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} = \frac{\frac{\omega}{\omega_0}}{\sqrt{1 + (\frac{\omega}{\omega_0})^2}} \quad \omega_0 = \frac{1}{RC}$$



• 带通电路.



($\omega_2 \gg \omega_1$).

实验内容: 1. 测低通滤波电路的幅频特性.

$R = 8k\Omega, C = 0.01\mu F, U_{in}$ 为正弦信号有效值 1V.

f / Hz	200	500	1k	1.5k	f_0	2k	3k	5k	10k	
U_2 / V	0.990	0.962	0.880	0.781	$0.707 U_1$	0.686	0.534	0.356	0.188	$f_0 = 1.89 \text{ kHz}$

2. 测高通电路幅频特性.

$R = 4k\Omega, C = 0.1\mu F$.

f / Hz	100	150	200	250	300	f_0	500	1k	5k	10k	
U_2 / V	0.244	0.352	0.448	0.530	0.599	$0.707 U_1$	0.777	0.920	0.987	0.990	$f_0 = 402 \text{ Hz}$

3. 测带通电路幅频特性. $R_1 = 4k\Omega, R_2 = 8k\Omega, C_1 = 0.1\mu F, C_2 = 0.01\mu F$.

f / Hz	50	100	200	$f_1 = 270 \text{ Hz}$	400	500	$f_0 = 865 \text{ Hz}$	1k	2k	$f_2 = 2.72 \text{ kHz}$	3k	5k	10k
U_2 / V	0.124	0.239	0.430	$0.707 U_{2max}$	0.644	0.695	U_{2max}	0.743	0.622	$0.707 U_{2max}$	0.497	0.338	0.180
				0.5287			0.7419			0.5287			





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思考题: 1. 不能. RC低通滤波电路若以电阻两端输出, 则输出频响特性与高通滤波电路相反.

2. 测量传输特性可获取电路幅频、相频特性, 验证截止频率、增益等设计参数. 保证了信号处理的效果.

数据计算与处理: 1. 低通: $f_{理} = \frac{1}{2\pi RC} = 1.990 \text{ kHz}$. $f_{实} = 1.89 \text{ kHz}$.

$$\text{相对误差} = \frac{0.1}{1.99} = 5.0\%$$

2. 高通: $f_{理} = \frac{1}{2\pi RC} = 398.1 \text{ Hz}$ $f_{实} = 402 \text{ Hz}$.

$$\text{相对误差} = \frac{3.9}{398.1} = 1.0\%$$

3. 带通: $f_{1理} = 398.1 \text{ Hz}$. $f_{1实} = 270 \text{ Hz}$. $\eta_1 = 32.2\%$

$$f_{2理} = 1.990 \text{ kHz} \quad f_{2实} = 2.72 \text{ kHz} \quad \eta_2 = 36.7\%$$

误差分析: ① 电表、导线存在电阻使 $f_{实}$ 偏小.

② 带通电路中, 未满足 $\omega_2 \gg \omega_1$, 从而实际与理论差别明显, 误差较大.



